ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PRACTICE EXAM 3 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Basic Principles of Computer Programming and Computer Literacy practice exam 3.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept any correct, equivalent code or algorithm. Do not penalise small syntax slips (e.g. a missing semicolon) more than once per question unless syntax is being tested. Give follow-through marks.

SECTION A

QUESTION 1

1.1 1.1.1 A ✓ 1.1.2 C ✓ 1.1.3 B ✓ 1.1.4 A ✓ 1.1.5 B ✓ 1.1.6 C ✓ 1.1.7 D ✓ 1.1.8 A ✓ 1.1.9 C ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 G ✓ 1.3.3 E ✓ 1.3.4 I ✓ 1.3.5 B ✓ 1.3.6 F ✓ 1.3.7 J ✓ 1.3.8 D ✓ 1.3.9 H ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 (a) input ✓ (b) output ✓ (c) storage ✓ (d) processing ✓ (4)

2.2 System software runs and manages the computer itself ✓, e.g. Windows, Linux, drivers, utilities ✓. Application software lets the user do a task ✓, e.g. Word, Excel, a web browser, the Arduino IDE ✓. (4)

2.3 Any THREE ✓✓✓: use folders with clear names; give files meaningful names (and dates / versions); save regularly; make backups (cloud / external drive); delete files you no longer need; do not save everything on the desktop. (3)

2.4.1 $=\text{SUM}(\text{B1:B4})$ ✓✓ $(=\text{B1}+\text{B2}+\text{B3}+\text{B4}\ \text{accepted})$ (2)

2.4.2 $=\text{AVERAGE}(\text{B1:B4})$ ✓ (1)

2.5 Any FOUR ✓✓✓✓: use strong, different passwords; do not open attachments or links from unknown senders (phishing); keep antivirus and software updated; do not share personal information or passwords; use secure (https) websites; log out on shared computers; think before posting on social media. (4)

[18]

QUESTION 3

3.1 A clear, step-by-step set of instructions ✓ that solves a problem and has a definite end ✓ (2)

3.2 Start / input length ✓; input width ✓; area = length × width ✓; display area / stop ✓ (4)

3.3 Oval Start ✓; parallelogram "input temperature" ✓; diamond "temperature > 40?" ✓; Yes → parallelogram "Fan ON" ✓; No → parallelogram "Fan OFF" ✓; both join to oval End, arrows correct ✓ (6)

3.4 a = 8 then b = 5 (8 − 3) then a = 3 (8 − 5) ✓✓; final a = 3, b = 5 ✓; the steps swap the values of a and b ✓ (4)

[16]

QUESTION 4

4.1 (a) includes the standard input/output library so that printf and scanf can be used ✓ (b) the main function where the program starts ✓ (c) mark the start and end of a block of code / the body of the function ✓ (d) ends main and tells the operating system the program finished successfully ✓ (4)

4.2 (a) int ✓ (b) float ✓ (c) char ✓ (3)

4.3 9 3 1 ✓✓✓ (3)

4.4 #include and main ✓; declare float variables ✓; prompt for length ✓ and read it with scanf("%f", &length) ✓; prompt and read width ✓; calculate area ✓; printf with %.2f ✓; return 0 / braces ✓, e.g. (8)

```
#include <stdio.h>

int main() {
  float length, width, area;
  printf("Enter the length: ");
  scanf("%f", &length);
  printf("Enter the width: ");
  scanf("%f", &width);
  area = length * width;
  printf("Area = %.2f\n", area);
  return 0;
}
```

4.5 Syntax error: breaks the rules of the language, so the program will not compile (e.g. a missing semicolon) ✓. Logic error: the program runs but gives the wrong answer (e.g. using + instead of ×) ✓. (2)

[20]

QUESTION 5

5.1 if with the correct condition mark >= 50 ✓; printf("Pass") ✓; else ✓; printf("Fail") ✓, e.g. (4)

```
if (mark >= 50) {
  printf("Pass");
} else {
  printf("Fail");
}
```

5.2 for with start 1 ✓, condition i <= 10 ✓, increment ✓, printf of i with a space ✓, e.g. (4)

```
for (int i = 1; i <= 10; i++) {
  printf("%d ", i);
}
```

5.3 1 3 9 ✓✓✓ (27 is not printed because the loop stops when i is 20 or more) (3)

5.4 "When green flag clicked" ✓; "forever" loop ✓ containing "move 10 steps" and "if on edge, bounce" ✓ (3)

5.5 Any TWO ✓✓: no syntax errors because blocks only fit together correctly; very visual and easy to understand; quick results that motivate learners; teaches logic (loops, decisions) before typing code. (2)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE